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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Employee Attendance Monitoring and Leave Management System

Submitted by

PRANEETH M

25BIT038

Under the Guidance of

DHILEEBAN A

Technical Trainer

Technical Department, KG College of Arts and Science

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DECLARATION

I hereby declare that the project entitled **"Employee Attendance Monitoring and Leave Management System"** submitted in partial fulfillment of the requirements for the course **JAVA PROGRAMMING** is a **record of my original work** carried out under the guidance of , Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

PRANEETH M

(Your Name)

CERTIFICATE

This is to certify that the project entitled **"Employee Attendance Monitoring and Leave Management System"** is a **bona fide work** carried out by **[PRANEETH M] [I-BScIT-.'A']**, [], **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[JAVA PROGRAMMING]**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: COIMBATORE

Date: 09.04.2026

Submitted for the Viva-Voce Examination held on **KG COLLAGE OF ARTS & SCIENCE**

Guide:
(Technical Trainer, KGCAS)

Head of the Department
(Technical Head, KGCAS)

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ABSTRACT

The **Employee Attendance Monitoring and Leave Management System** is a Java-based application developed to efficiently manage employee attendance records and handle leave requests in an organized manner. In many organizations, attendance and leave details are often recorded manually or maintained across different systems, which can lead to issues such as data duplication, errors in attendance tracking, delays in leave approvals, and difficulty in maintaining accurate records.

This project aims to simplify attendance and leave management by providing an automated system that allows users to store, retrieve, update, and manage employee attendance and leave details easily. The system uses a menu-driven interface that enables users to mark attendance, apply for leave, and view records efficiently, ensuring better organization and quick access to information.

The application is developed using Core Java programming concepts such as classes, objects, methods, loops, conditional statements, and file handling. It is implemented as a console-based application and can be further enhanced using Java Swing to provide a graphical user interface for improved user experience.

The main objective of the system is to improve operational efficiency, reduce manual effort, and ensure accurate tracking of employee attendance and leave records. The system provides key features such as adding employee details, marking daily attendance, applying for leave, updating records, and viewing reports, making it suitable for small organizations, institutions, and academic projects.

This project demonstrates the practical implementation of object-oriented programming in Java and serves as a foundation for developing advanced systems with features like database integration, biometric attendance, real-time monitoring, cloud storage, and enhanced security in the future.

1.INTRODUCTION

In today's digital world, managing employee attendance and leave efficiently is very important for organizations. The **Employee Attendance Monitoring and Leave Management System** is a software application developed to store, organize, and manage employee attendance records and leave details in a structured and user-friendly manner.

This project is designed as a simple application that allows users to perform operations such as adding new employees, marking daily attendance, applying for leave, updating records, and deleting unnecessary data. It helps maintain all employee-related information in one place, avoiding common problems of manual record keeping such as data loss, duplication, delays, and difficulty in tracking attendance and leave status.

The system can be implemented as a console-based application using programming languages like Java, Python, or C++. It uses basic programming concepts such as arrays or collections, loops, conditional statements, and functions to perform various operations related to attendance monitoring and leave management.

The main objective of this project is to reduce manual effort and provide quick and accurate access to employee attendance and leave details. The tracking feature allows users to monitor attendance status, while the update feature ensures that records can be modified whenever required.

This project is useful for students, small organizations, and institutions to manage employee attendance efficiently. It also demonstrates the practical application of programming concepts and serves as a foundation for developing advanced systems with features like database integration, graphical user interface, real-time tracking, and cloud storage.

1.1 Objective

The main objective of the *Employee Attendance Monitoring and Leave Management System* is to efficiently manage employee attendance records and leave requests. This system helps reduce manual work, improve accuracy, and maintain organized records.

1.2 Overview

This system is designed to track employee attendance and manage leave applications in an organization. It allows adding employee details, marking daily attendance, applying for leave, and viewing leave status. The system is developed using Java without a database, making it simple and suitable for academic purposes.

1.3 Features

- Add new employees
- View employee details
- Mark daily attendance
- Apply for leave
- View leave records
- Simple and user-friendly interfac
-

2. Program Analysis

2.1 Problem Definition

Manual attendance tracking and leave management systems are time-consuming and prone to errors. Maintaining records manually can lead to data loss and inefficiency.

2.2 Project Overview

This project provides a computerized solution using Java. It stores employee data in memory and allows users to manage attendance and leave records through a menu-driven program.

2.3 Scopes and Limitations

Scope:

- Useful for small organizations
- Easy to use and implement
- Can be extended with GUI or database

Limitations:

- Data is not permanently stored (in current version)
- No user authentication
- No advanced reporting system

3.SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The **Employee Attendance Monitoring and Leave Management System** is designed as a menu-driven application where users can select food items, enter quantities, and generate a final bill automatically.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system.

System design defines the structure and working of the system

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3.1 Modules

- **Employee Module:** Add and view employee details
- **Attendance Module:** Mark daily attendance
- **Leave Module:** Apply and view leave status
- **Main Module:** Menu-driven control system

4.METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the **Employee Attendance Monitoring and Leave Management System** History a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

• Problem Analysis
• System Design
• Program Development
• Testing and Validation
• Result and Documentation

The **Employee Attendance Monitoring and Leave Management System** follows a structured development approach.

4.1 Problem Analysis

Understanding the need for an automated system to replace manual record keeping.

4.2 System Design

The system is designed using a modular approach with separate classes for Employee and Leave management.

4.3 Program Development

The system is developed using Java programming language with:

- **Classes and Objects**
- **ArrayList for data storage**
- **Scanner for user input**

4.4 Testing and Validation

- **Tested each module individually**

- **Checked for input errors**
- **Verified correct data display**

4.5 Result and Documentation

The system successfully manages employee attendance and leave records through a console interface.

4.6 Algorithm

Step 1: Start program
Step 2: Display menu options
Step 3: User selects option
Step 4: Perform selected operation

- **Add employee**
- **View employee**
- **Mark attendance**
- **Apply leave**
- **View leave**

Step 5: Repeat until exit
Step 6: Stop

4.7 Flowchart (Text Representation)

Start

↓

Display Menu

↓

Enter Choice

↓

Add Employee	Mark Attendance	Apply Leave

↓

View Data

↓

Exit?

↓

Yes → End

No → Repeat

5.SOURCE CODE

```
import java.io.*;
import java.time.LocalDate;
import java.util.*;

class Employee {
    int id;
    String name;
    String email;

    Employee(int id, String name, String email) {
        this.id = id;
        this.name = name;
        this.email = email;
    }
}

class Leave {
    int empId;
    String fromDate;
    String toDate;
    String reason;
    String status;

    Leave(int empId, String fromDate, String toDate, String reason,
String status) {
```

```
        this.empId = empId;
        this.fromDate = fromDate;
        this.toDate = toDate;
        this.reason = reason;
        this.status = status;
    }
}

public class Main {
    static List<Employee> employees = new ArrayList<>();
    static List<String> attendance = new ArrayList<>();
    static List<Leave> leaves = new ArrayList<>();

    static int empIdCounter = 1;

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        while (true) {
            System.out.println("\n1. Add Employee");
            System.out.println("2. View Employees");
            System.out.println("3. Mark Attendance");
            System.out.println("4. Apply Leave");
            System.out.println("5. View Leaves");
            System.out.println("6. Exit");
            System.out.print("Choose: ");

            int choice = sc.nextInt();
            sc.nextLine();
```

```
            switch (choice) {
                case 1 -> addEmployee(sc);
                case 2 -> viewEmployees();
                case 3 -> markAttendance(sc);
                case 4 -> applyLeave(sc);
                case 5 -> viewLeaves();
                case 6 -> System.exit(0);
            }
        }
    }

    static void addEmployee(Scanner sc) {
        System.out.print("Name: ");
        String name = sc.nextLine();
        System.out.print("Email: ");
        String email = sc.nextLine();

        Employee emp = new Employee(empIdCounter++, name, email);
        employees.add(emp);

        System.out.println("Employee added!");
    }

    static void viewEmployees() {
        for (Employee e : employees) {
            System.out.println(e.id + " - " + e.name + " - " +
e.email);
        }
```

```
    }

    static void markAttendance(Scanner sc) {
        System.out.print("Employee ID: ");
        int id = sc.nextInt();

        String record = "EmpID: " + id + " Date: " + LocalDate.now() +
" Status: Present";
        attendance.add(record);

        System.out.println("Attendance marked!");
    }

    static void applyLeave(Scanner sc) {
        System.out.print("Employee ID: ");
        int id = sc.nextInt();
        sc.nextLine();

        System.out.print("From Date (YYYY-MM-DD): ");
        String from = sc.nextLine();
        System.out.print("To Date (YYYY-MM-DD): ");
        String to = sc.nextLine();
        System.out.print("Reason: ");
        String reason = sc.nextLine();

        Leave leave = new Leave(id, from, to, reason, "Pending");
        leaves.add(leave);

        System.out.println("Leave applied!");
```

```
    }

    static void viewLeaves() {
        for (Leave l : leaves) {
            System.out.println("EmpID: " + l.empId +
                    " | " + l.fromDate + " to " + l.toDate +
                    " | " + l.reason +
                    " | " + l.status);
        }
    }
}

/*
FEATURES:
- No SQL / No Database
- Uses ArrayList (memory-based)
- Simple and easy for mini project

OPTIONAL IMPROVEMENT:
- Save data into file using FileWriter
- Load data using FileReader
*/
```

6.INPUT AND OUTPUT SCREENS

```
1. Add Employee
2. View Employees
3. Mark Attendance
4. Apply Leave
5. View Leaves
6. Exit
Choose: 1
Name: praneeth
Email: praneeth32@gmail.com
Employee added!
```

```
1. Add Employee
2. View Employees
3. Mark Attendance
4. Apply Leave
5. View Leaves
6. Exit
Choose: 2
1 - praneeth  - praneeth32@gmail.com
```

```
1. Add Employee
2. View Employees
3. Mark Attendance
4. Apply Leave
5. View Leaves
6. Exit
Choose: 3
Employee ID: 1
Attendance marked!
```

```
1. Add Employee
2. View Employees
3. Mark Attendance
4. Apply Leave
5. View Leaves
6. Exit
Choose: 4
Employee ID: 1
From Date (YYYY-MM-DD): 2026-04-17
To Date (YYYY-MM-DD): 2026-04-20
Reason: function
Leave applied!
```

```
1. Add Employee
2. View Employees
3. Mark Attendance
4. Apply Leave
5. View Leaves
6. Exit
Choose: 5
EmpID: 1 | 2007-10-19 to 2026-10-15 | function | Pending
EmpID: 1 | 2026-04-17 to 2026-04-20 | function | Pending
```

```
1. Add Employee
2. View Employees
3. Mark Attendance
4. Apply Leave
5. View Leaves
6. Exit
Choose: 6
```

7.DISCUSSION & CONCLUSION

7.1 Discussion

The system simplifies attendance and leave management. It reduces manual effort and improves efficiency. However, it can be improved by adding database support and GUI.

7.2 Conclusion

The project successfully demonstrates how a basic employee management system can be implemented using Java. It is efficient for small-scale applications and serves as a foundation for more advanced systems.

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8.REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

• https://www.oracle.com/java
• https://www.geeksforgeeks.org
• https://www.w3schools.com
• https://www.javatpoint.com